

Secretary Problem Research

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Contents

1	Abstract Error Secretary Problem with Examination	2
1.1	Problem Formulation	2
1.2	Algorithmic Representations	3
2	Relaxation of the abstract error SP problem	4
2.1	Further relaxed model	5
2.2	relaxed model	6
2.2.1	Calculation 2 (more compact, and considers edge case) . .	7
2.3	Lower Bound of $P^{\max}$	7
2.4	Upper Bound of $P^{\max}$	8
2.5	Responses	9
2.5.1	Response to exercise 1	9
2.5.2	Response to exercise 2	9
2.6	Calculation with error involved	9
2.7	Another Approach to Estimate $P^{\max}$	11
3	DP solution to Simple Problem	12
4	Relaxed Problem with Error	13
5	Noisy SP with examine	16
5.1	Problem Construction from R.V.s	16
5.2	Problem Construction from permutation group	16
6	Partially Adversarial SP	18
6.1	Problem formulation	18
6.2	Preliminaries	18
6.3	Case without examination	19
6.3.1	Weak adversarial, do not know anything	19
6.3.2	Weak adversarial, know algorithm	19
6.3.3	Strong adversarial, knows τ and σ	20
6.3.4	Strong adversarial, knows τ , σ and $\mathbf{r}$	20
6.4	Lower Bound for an algorithm	20
6.5	Upperbound for the problem	24
6.5.1	some guesses	24
6.5.2	An alternative thought	24

Lecture 1: abstract error Secretary

Sun 30 Jan 2022 15:42

1 Abstract Error Secretary Problem with Examination

1.1 Problem Formulation

n items arrive in random order. Let i denote the item that arrives in the round i . The items are assigned absolute ranks r_i . r is totally ordered wrt i . Define d_i to be the player's decision at round i . It equals one of the following: Accept, Decline, Examine

- $d_i = \text{Accept} \Rightarrow \text{Terminates}$
- $d_i = \text{Accept} \wedge \forall j \in [n], r_j < r_i \Leftrightarrow \text{Wins}$

Define O_i to be the items that the player has **observed** by round i , and E_i to be the items that player has **examined** by round i .

- $O_1 = [1], E_1 = \emptyset$
- $E_i \subseteq O_i$
- $d_i = \text{Decline} \Rightarrow O_{i+1} = O_i \cup \{i+1\}, E_{i+1} = E_i$
- $d_i = \text{Examine} \Rightarrow O_{i+1} = O_i, E_{i+1} = E_i \cup \{i\}$

Definition 1 (Indistinguishable Items). $i \approx j$ means that i and j are in-distinguishable given that neither of them have been examined.

- (reflexivity) $i \approx i$
- (symmetry) $i \approx j \Leftrightarrow j \approx i$
- (downward closure) $i \approx j, r_i < r_j \Rightarrow \forall k, l [r_k, r_l \in [r_i, r_j] \Rightarrow r_k \approx r_l]$

Define $N_i := \{j : j \approx i\}$ to be the *neighbour* of i . Note that $1 \leq |N_i| \leq n$. Define Θ_i to be the relations known to the player by round i .

- Θ_k defines a partial order $\leq_k$ on O_k , such that $i \leq_k j \Leftrightarrow (i, j) \in \Theta_k$
- $\Theta_1 = \{(1, 1)\}$
- $i \leq_k j \Rightarrow i \leq_{k+1} j$
- $\forall i, j \in O_k [i \not\approx j \Rightarrow i \leq_k j \vee j \leq_k i]$
- (E_k totally ordered) $\forall i, j \in E_k [i \leq_k j \vee j \leq_k i]$.

Summary: For an incoming item i , upon first observation, the player can compare r_i and r_j for $\forall j \in O_i \setminus N_i$. Upon the second observation, the player can compare the examined item with **previously examined** items (which can be in N_i).

1.2 Algorithmic Representations

Algorithm 1 Player's algorithm for the problem

```
1: for  $i \in [n]$  do
2:    $\alpha \leftarrow$  the current item to be decided
3:   if  $i < r$  then ▷ Phase 1
4:     if  $\exists \beta \in O_i. \alpha <_i \beta$  then ▷  $\alpha$  is not a maximum
5:       Decline this item
6:     else if  $\alpha \in E_i$  then
7:       Decline this item
8:     else
9:       Examine this item
10:    end if
11:  else if  $i < n$  then ▷ Phase 2
12:    if  $\exists \beta \in O_i. \alpha <_i \beta$  then
13:      Decline this item
14:    else
15:      if  $\forall \beta \in O_i. \beta \leq_i \alpha$  then ▷  $\alpha$  is the absolute maximum
16:        Accept this item
17:      else if  $\alpha \in E_i$  then
18:        Decline this item
19:      else
20:        Examine this item
21:      end if
22:    end if
23:  else
24:    Accept this item
25:  end if
26: end for
```

Algorithm 2 Game Process

```

1:  $[n], r_i$  and  $\approx$  relation are given according to definition
2:  $O_1 = \{1\}$ 
3:  $E_1 = \emptyset$ 
4:  $\alpha_1 = 1$ 
5:  $\Theta_1 = \emptyset$ 
6: for  $i \in [n]$  do
7:    $d_i \leftarrow$  player's decision
8:   if  $d_i = \text{Accept}$  then
9:     Game terminates with reward  $v(\alpha_i)$ 
10:  else if  $d_i = \text{Decline}$  then
11:     $O_{i+1} \leftarrow O_i \cup \{i+1\}$ 
12:     $E_{i+1} \leftarrow E_i$ 
13:     $\alpha_{i+1} \leftarrow i+1$ 
14:  else if  $d_i = \text{Examine}$  then
15:     $O_{i+1} \leftarrow O_i$ 
16:     $E_{i+1} \leftarrow E_i \cup \{i\}$ 
17:     $\alpha_{i+1} \leftarrow \alpha$ 
18:  end if
19:   $\Theta_{i+1} \leftarrow \{(\beta, \gamma) \in O_{i+1}^2 : r_\beta \leq r_\gamma, \beta \not\approx \gamma\} \cup \{(\beta, \gamma) \in E_{i+1}^2 : r_\beta \leq r_\gamma\}$ 
20: end for

```

2 Relaxation of the abstract error SP problem

In the previous definition, i and j need to both have been examined, for the player to compare them precisely. Now we relax the problem, needing only one examination between i and j to obtain precise comparison.

We give the expected value of ALG on the relaxed model. Let α denote the selected item.

$$\mathbb{E}[\text{OPT}] = m \quad (1)$$

$$\mathbb{E}[\text{ALG}] = \sum_{t=1}^m t \Pr[v(\alpha) = t] \quad (2)$$

$$\Pr[v(\alpha) = t] = \sum_{i=r}^n \Pr[\alpha = i | v(i) = t] \Pr[v(i) = t] \quad (3)$$

$$+ \Pr[\text{reach } n \cap \text{not accept } n] \Pr[v(n) = t] \quad (4)$$

$$+ \Pr[\text{reach } n-1 \cap \text{ED } n-1] \Pr[v(n-1) = t] \quad (5)$$

The two added terms are cases where the items are exhausted. If n is reached then it must be selected. If $n-1$ is examined then it must be selected also.

$$\Pr[v(i) = t] = 1/m, \forall i \forall t \quad (6)$$

$$\Pr[\alpha = i | v(i) = t] = \Pr\left[\text{reach } i \mid \max_{j < r} v(j) \leq t\right] \Pr\left[\max_{j < r} v(j) \leq t\right] \quad (7)$$

$$= \sum_{s=1}^t \Pr\left[\text{reach } i \mid \max_{j < r} v(j) = s\right] \Pr\left[\max_{j < r} v(j) = s\right] \quad (8)$$

$$\Pr[\text{reach } n \cap \text{not accept } n] = \sum_{s=t+1}^m \Pr\left[\text{reach } n \mid \max_{j < r} v(j) = s\right] \Pr\left[\max_{j < r} v(j) = s\right] \quad (9)$$

$$\Pr[\text{reach } n-1 \cap \text{ED } n-1] = \sum_{s=t+1}^m \mathbf{1}_{s \leq t+\delta} \Pr\left[\text{reach } n-1 \mid \max_{j < r} v(j) = s\right] \Pr\left[\max_{j < r} v(j) = s\right] \quad (10)$$

$$\Pr\left[\max_{j < r} v(j) = s\right] = \Pr\left[\max_{j < r} v(j) \leq s\right] - \Pr\left[\max_{j < r} v(j) \leq s-1\right] \quad (11)$$

$$= \left(\frac{s}{m}\right)^{r-1} - \left(\frac{s-1}{m}\right)^{r-1} \quad (12)$$

$$\Pr\left[\text{reach } i \mid \max_{j < r} v(j) = s\right] \quad (13)$$

$$= \sum_{(a_1 \dots a_k) \in \{1,2\}^k, \sum a_j = i-r} \prod_{j=1}^k \mathbf{1}_{\delta < t} \left(\mathbf{1}_{a_j=1} \frac{s-\delta-1}{m} + \mathbf{1}_{a_j=2} \frac{\delta}{m} \right) + \mathbf{1}_{\delta \geq t} \frac{t-1}{m} \quad (14)$$

$$= \sum_{x=0}^{\lfloor \frac{i-r}{2} \rfloor} \left(\mathbf{1}_{\delta < t} \left(\frac{\delta}{m} \right)^x \left(\frac{s-\delta-1}{m} \right)^{i-r-2x} + \mathbf{1}_{\delta \geq t} \left(\frac{t-1}{m} \right)^x 0^{i-r-2x} \right) \times \frac{(i-r-x)!}{x!(i-r-2x)!} \quad (15)$$

Explanation: There are $i-r$ steps to go from r to i . Each decline will go 1 step, and each examine+decline will go 2 steps. We calculate the probability to take each possible path and then sum them up. In the second equation, x is the number of items examined going from r to i .

Lecture 2: Relaxed SP

Sun 30 Jan 2022 15:50

2.1 Further relaxed model

Now we consider a relaxed version of classical SP. n items have values i.i.d. uniformly distributed in $[0, 1]$. We want to observe sequentially and choose the best item.

We explore the best result possible when we use a simple **thresholding scheme**: pick the first item with value $\geq T$.

2.2 relaxed model

calculation 1 (discussed in meeting with zijie)

$$\begin{aligned}
 & \sum_{i=1}^n \Pr[\text{accept } i \cap i \text{ best}] \\
 &= \sum_{i=1}^n \Pr[\text{accept } i | i \text{ best}] \Pr[i \text{ best}] \\
 &= \frac{1}{n} \sum_{i=1}^n \Pr[X_1, \dots, X_{i-1} < T \cap X_i \geq T | i \text{ best}] \\
 &= \frac{1}{n} \sum_{i=1}^n \Pr[X_1, \dots, X_{i-1} < T | X_i \geq T \cap i \text{ best}] \Pr[X_i \geq T | i \text{ best}]
 \end{aligned}$$

Note that $\Pr[X_i \geq T | i \text{ best}] = 1 - F_{X_{\max}}(T) = 1 - T^n$. Also,

$$\begin{aligned}
 & \Pr[X_1, \dots, X_{i-1} < T | X_i \geq T \cap i \text{ best}] \\
 &= \int_T^1 \Pr[X_1, \dots, X_{i-1} < T | i \text{ best} \cap X_i = x] f_X(x | X_i \geq T \cap i \text{ best}) dx \\
 &= \int_T^1 \Pr[X_1, \dots, X_{i-1} < T | X_1, \dots, X_{i-1} < x] f_{X_{\max}}(x | X_{\max} \geq T) dx \\
 &= \int_T^1 \left(\frac{T}{x}\right)^{i-1} \frac{nx^{n-1}}{1 - T^n} dx \\
 &= \frac{nT^{i-1}}{1 - T^n} \int_T^1 x^{n-i} dx \\
 &= \frac{nT^{i-1}}{1 - T^n} \frac{1 - T^{n-i+1}}{n - i + 1}
 \end{aligned}$$

Combining all terms,

$$\begin{aligned}
 & \sum_{i=1}^n \Pr[\text{accept } i \cap i \text{ best}] \\
 &= \frac{1}{n} \sum_{i=1}^n \frac{nT^{i-1}}{1 - T^n} \frac{x^{n-i+1}}{n - i + 1} (1 - T^n) \\
 &= \sum_{i=1}^n T^{i-1} \frac{1 - T^{n-i+1}}{n - i + 1}
 \end{aligned}$$

2.2.1 Calculation 2 (more compact, and considers edge case)

If the player reaches X_n , then X_n must be accepted.

$$\begin{aligned}
 \Pr[X_{max} \text{ selected}] &= \sum_{i=1}^{n-1} \Pr[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_i \text{ is max}] \\
 &\quad + \Pr[X_1, \dots, X_{n-1} < T \cap X_n \text{ max}] \\
 &= \sum_{i=1}^{n-1} \int_0^1 \Pr[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_{i+1}, \dots, X_n \leq X_i | X_i = x] f_{\text{Unif}(0,1)}(x) dx \\
 &\quad + \Pr[X_n \text{ max}] \int_0^1 \Pr[X_1, \dots, X_{n-1} < T | X_n = x \text{ is max}] f_{\text{max Unif}(0,1)}(x) dx \\
 &= \sum_{i=1}^{n-1} \int_T^1 \Pr[X_1, \dots, X_{i-1} < T \cap X_{i+1}, \dots, X_n \leq x] f_{\text{Unif}(0,1)}(x) dx \\
 &\quad + \frac{1}{n} \int_0^1 \Pr[X_i < T | X_i < x]^{n-1} f_{\text{max Unif}(0,1)}(x) dx \\
 &= \sum_{i=1}^{n-1} \int_T^1 T^{i-1} x^{n-i} dx + \int_0^T x^{n-1} dx + \int_T^1 x^{n-1} \left(\frac{T}{x}\right)^{n-1} dx \\
 &= \sum_{i=1}^{n-1} \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}) + \frac{1}{n} T^n + T^{n-1} (1 - T)
 \end{aligned}$$

2.3 Lower Bound of $P^{\max}$

This section is authored by Zijie Zhou.

In this section, we provide a lower bound of $P^{\max}$, where

$$P^{\max} = \max_{T \in [0,1]} \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}).$$

(Since we are interested in asymptotic behavior as $n \rightarrow \infty$, the difference in the last term can be safely ignored.)

The optimal T^* can be solved by the optimization problem. Let's show the lower bound of $P^{\max}(T^*)$. Let $T = \frac{n}{n+1}$, then we have

$$\begin{aligned}
 P^{\max}(T^*) &\geq P^{\max}(T) \\
 &= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}) \\
 &= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} - \sum_{i=1}^n \frac{1}{n-i+1} T^n \\
 &= T^n \sum_{i=1}^n \frac{1}{i} (T^{-i} - 1)
 \end{aligned}$$

We take the limit $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} T^n \sum_{i=1}^n \frac{1}{i} (T^{-i} - 1) = \frac{1}{e} \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{i} \left(\left(1 + \frac{1}{n}\right)^i - 1 \right)$$

Observe that

$$\left(1 + \frac{1}{n}\right)^i - 1 = \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k}.$$

Then, we take the sum on i , we have

$$\sum_{i=1}^n \frac{1}{i} \left(1 + \frac{1}{n}\right)^i - 1 = \sum_{i=1}^n \frac{1}{i} \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k}$$

By Fubini Theorem, we have

$$\sum_{i=1}^n \frac{1}{i} \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k} = \sum_{k=1}^n \sum_{i=k}^n \frac{1}{i} \binom{i}{k} \frac{1}{n^k}$$

We have

$$\frac{1}{i} \binom{i}{k} = \frac{(i-1)(i-2)\dots(i-k+1)}{k!}$$

The nominator is a polynomial with order $k-1$ with respect to i .

Show that $\sum_{i=1}^n i^k = \frac{1}{k+1} n^{k+1} + O(n^k)$

By this property, we have

$$\lim_{n \rightarrow \infty} \sum_{i=k}^n \frac{1}{i} \binom{i}{k} \frac{1}{n^k} = \frac{1}{k \cdot k!} \frac{n^k}{n^k} = \frac{1}{k \cdot k!}.$$

Finally, we have the lower bound of $P^{\max}$ is

$$\frac{1}{e} \sum_{k=1}^n \frac{1}{k \cdot k!} \approx 0.485.$$

2.4 Upper Bound of $P^{\max}$

Consider there are only two secretaries in the arriving sequence. We have $X_1 \sim \text{Unif}(0, 1)$, $X_2 \sim \text{Unif}(0, 1)$.

Show that the upper bound of $P^{\max}$ among all deterministic algorithms is $\frac{1}{2}$

Hint: Any deterministic algorithm can only choose X_1 or X_2 . If $X_1 = x$, what will happen? Then take integral. Another method is that think about a coupling random arrival instance $Y_1 = X_1$, $Y_2 = \text{Unif}(0, 1)$. Y obviously follows the arrival process distribution. When the first customer arrives, any algorithm cannot distinguish the arrival instance is X or Y . Then for the first customer, if the algorithm accepts him, ..., if not accept,....

Show that the upper bound of $P^{\max}$ among all randomized algorithms is $\frac{1}{2}$

Hint: Yao's principle. (or set a probability distribution on X_1 and X_2).

Lecture 3: response to exercises

Mon 07 Feb 2022 00:24

2.5 Responses

2.5.1 Response to exercise 1

Proof. Use induction on n . The base case is trivial:

$$1 = \frac{1}{k+1} + O(1).$$

Now suppose the property holds for n .

$$\begin{aligned} \sum_{i=1}^{n+1} i^k &= (n+1)^k + \sum_{i=1}^n i^k = (n+1)^k + \frac{1}{k+1} n^{k+1} + O(n^k) \\ \frac{1}{k+1} (n+1)^{k+1} &= \frac{1}{k+1} n^{k+1} + O(n^k). \end{aligned}$$

By considering $(n+1)^k = O(n^k)$ and $O(n^k) \pm O(n^k) = O(n^k)$, the difference of the right hand side of the above equations is $O(n^k)$. The left hand side shows the property for $n+1$. $\square$

2.5.2 Response to exercise 2

Optimal P for a deterministic algorithm is 0.75.

Proof. ALG is completely characterized by a function $\alpha : [0, 1] \rightarrow [0, 1]$, where $\alpha(x)$ represents the probability of accepting X_1 after an observation $X_1 = x$.

$$\begin{aligned} \Pr[\text{win}] &= \int_0^1 \alpha \Pr[X_2 < x] + (1 - \alpha) \Pr[X_2 > x] f(x) dx \\ &= \int_0^1 \alpha x + (1 - \alpha)(1 - x) dx \\ &= \int_0^1 2\alpha x - \alpha dx + \frac{1}{2} \\ &\leq \max_{\alpha} \int_0^1 2\alpha x - \alpha dx + \frac{1}{2}. \end{aligned}$$

ALG is deterministic iff $\alpha(x) \in \{0, 1\}$. If α takes 0, the integral is zero. If α takes 1, $2\alpha x - \alpha > 0$ iff $x > \frac{1}{2}$. Hence to maximize the integral we take

$$\alpha(x) = \begin{cases} 1 & x \geq \frac{1}{2} \\ 0 & x < \frac{1}{2} \end{cases}, \text{ which gives an upper bound of } \frac{3}{4}. \quad \square$$

Lecture 4: Relaxed SP with error

Tue 01 Feb 2022 00:06

2.6 Calculation with error involved

The weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. The noisy observations $\tilde{X}_1, \dots, \tilde{X}_n \sim \text{Unif}(x_i - \delta, x_i + \delta)$.

The strategy is to introduce thresholding parameters R and $T \in (0, 1)$. We examine the upcoming item i if $\tilde{x}_i > R$, and accept the item if $x_i > T$. We still want to find $\Pr[X_{max} \text{ is selected}]$ given particular R, T, n, δ .

$$\begin{aligned} \Pr[X_{max} \text{ selected}] &= \sum_{i=1}^{n-2} \int_0^1 \Pr[X_i \text{ best} \cap i \text{ selected} | X_i = x] f(x) dx \\ &\quad + \Pr[\text{reach } n-1 \cap X_{n-1} \text{ best} \cap \tilde{X}_{n-1} > R] \\ &\quad + \Pr[\text{reach } n \cap X_n \text{ best}] \end{aligned} \quad (16)$$

For all $i \leq i-2$:

$$\begin{aligned} &\Pr[X_i \text{ best} \cap i \text{ selected} | X_i = x] \\ &= \Pr[\text{reach } i \cap \tilde{X}_i > R \cap X_i > T \cap X_i \text{ max} | X_i = x] \\ &= \Pr[\text{Unif}(x - \delta, x + \delta) > R] \mathbf{1}(x > T) \Pr[\text{reach } i \cap X_i \text{ max} | X_i = x] \end{aligned}$$

For all $i \leq i-2$:

$$\begin{aligned} &\Pr[\text{reach } i \cap X_i \text{ max} | X_i = x] \\ &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \Pr[\text{reach } i \cap X_i \text{ max} \cap k \text{ examines before reaching } i | X_i = x] \\ &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \Pr[\text{reach } i \text{ with } k \text{ examines}] \Pr[X_i \text{ max} | \text{reach } i \text{ with } k \text{ examines} \cap X_i = x] \\ &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \underbrace{\frac{(i-1-k)!}{k!(i-1-2k)!}}_{\text{ways to reach } i} \Pr[\text{ED}]^k \Pr[\text{D}]^{i-1-2k} \underbrace{F(x|X < T)^k}_{\text{examined}} \underbrace{F(x)^k}_{\text{omitted}} \underbrace{F(x|\tilde{X} < R)^{i-1-2k}}_{\text{declined directly}} \underbrace{F(x)^{n-i}}_{\text{upcoming}} \end{aligned}$$

where

$$\begin{aligned} \Pr[\text{ED}] &= \frac{1}{2\delta} \int_0^T \int_{y-\delta}^{y+\delta} \mathbf{1}(z > R) dz dy = \text{const} \\ \Pr[\text{D}] &= \frac{1}{2\delta} \int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy = \text{const} \end{aligned}$$

and for $0 \leq x \leq 1$:

$$\begin{aligned} F(x|X < T) &= \begin{cases} \frac{x}{T} & 0 \leq x < T \\ 1 & x \geq T \end{cases} \\ F(x|\tilde{X} < R) &= \frac{\int_0^x \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy}{\int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy} \\ F(x) &= x. \end{aligned}$$

2.7 Another Approach to Estimate $P^{\max}$

This section is mostly credited to Zijie Zhou

To simplify the writing, we write Error Win as the event that we find the best secretary under the model with error. We write Original Win as the event that we find the best secretary under the original model.

$$\begin{aligned}\Pr(\text{Error Win}) &= \Pr(\text{Error Win}|\text{Original Win}) \Pr(\text{Original Win}) \\ &\quad + \Pr(\text{Error Win}|\text{Original Lose}) \Pr(\text{Original Lose}) \\ &\geq 0.485 \Pr(\text{Error Win}|\text{Original Win}).\end{aligned}$$

Optional exercise: Is there any method to bound $\Pr(\text{Error Win}|\text{Original Lose})$.

Next, we bound the probability of Error win conditional on Original Win. We start with defining two events: we let $\mathcal{E}_x = \max X_i = x$. We let $\mathcal{F}_y$ be that the secretary who arrives in the last period before the max secretary arrives. Because the arriving process is i.i.d, it is easy to see that $\mathcal{F}_y \sim \text{Unif}(0, 1)$.

Exercise1: Find the distribution of $\mathcal{E}_x$.

Hint: See Beta distribution.

We have

$$\Pr(\text{Error Win}|\text{Original Win}) = \int_{x,y} \Pr(\text{Error Win}|\text{Original Win}, \mathcal{E}_x, \mathcal{F}_y) f(x) dx dy.$$

We construct two arriving instances as follows: let I_1 contains n i.i.d. $\text{Unif}(0, 1)$ r.v. In addition, the maximal realization is x . The last r.r. before the max one has realization y . Let I_2 be a copy of I_1 . Conditional on Original win, there are two cases that we do not win on the same instance:

- The max x has observation $\hat{x} < T$, then we do not examine the best secretary.
- The one before max, y , has observation $\hat{y} \geq T$, then the best secretary is skipped. (Notice that here we do not need to consider the second last before x and all the previous r.v., because we are providing the lower bound.)

Therefore, we have

$$\Pr(\text{Error lose}|\text{Original Win}, \mathcal{E}_x, \mathcal{F}_y) \leq \Pr(\text{Case 1} \cup \text{Case 2}) \leq \Pr(\text{Case 1}) + \Pr(\text{Case 2}).$$

Exercise 2: Find the probability of Case 1 and Case 2 by considering that the observation cannot exceed 1 or below 0.

All calculation were based on the assumption that $\delta < T < 1 - \delta$.

$$F(x) = \begin{cases} 1 & x < T - \delta \\ 0 & x > T + \delta \\ \frac{x + \delta - T}{2\delta} & \text{otherwise} \end{cases}$$

$$\Pr(\text{Case 1}) = F(x)$$

$$\Pr(\text{Case 2}) = 1 - F(y)$$

Finally, we have

$$\begin{aligned} \Pr(\text{Error Win}) &\geq 0.485 \Pr(\text{Error Win} | \text{Original Win}) \\ &\geq 0.485 \int_{x,y} (1 - \Pr(\text{Case 1}) - \Pr(\text{Case 2})) f(x) dx dy. \end{aligned}$$

Please complete the final estimation. Write it as a function of δ , if the integral is hard to calculate, run numerical experiments for different δ and see what the value of the integral is.

Evaluated Integral:

$$\frac{((-2n-1)\delta - T)(T - \delta)^n + ((-2n-1)\delta + T)(\delta + T)^n + 2T\delta(n+1)}{2(n+1)\delta}$$

$$\xrightarrow{n \rightarrow \infty} T$$

$$\begin{aligned} \Pr[\text{Error Win} | \text{Original Lose}] &= \Pr[\text{reach } i | \exists j < i, X_j > T] \Pr[X_{\max} > T | \text{Original Lose}] \\ &\leq \Pr[j - 1 \text{ examine}] \Pr[X_{\max} > T | \text{Original Lose}] \end{aligned}$$

Lecture 5: Properties of the DP solution to the relaxed problem with and without error

Wed 09 Mar 2022 23:23

3 DP solution to Simple Problem

We begin with a relaxed problem of best selection with no error. Let $[x, n]$ denote the initial game state where we have an upcoming item with value x and n un-observed items after the current item.

If the player accepts the first item, he wins iff all other items are less than the first item. Hence $\Pr[\text{win} | \text{accept at } [x, n]] = x^n$. If the player declines x , he wins iff he wins the sub-game $[x_{n-1}, n-1]$, and $\max(x_{n-1}, \dots, x_1) < x$. We denote the new game state by $[x_{n-1}, n-1, x]$.

Let $V(x, n, x^*) := \Pr[\text{win starting from } [x, n] \cap \max(x, x_n, \dots, x_0) > x^*]$. From principle of dynamic programming, the optimal policy can be yielded from the

following system of equations:

$$\begin{aligned} V(x, 0, x^*) &= \mathbf{1}(x > x^*) \\ V(x, n, x^*) &= \begin{cases} \max(x^n, EV(\cdot, n-1, x)) & x \geq x^* \\ EV(\cdot, n-1, x^*) & x < x^* \end{cases} \end{aligned}$$

Our first observation is that a thresholding scheme can be optimal.

Theorem 1. There exists a sequence $T_n \subset \mathbb{R}$ such that $x^n > EV(\cdot, n-1, x) \Leftrightarrow x > T_n$ for all $n \in \mathbb{N}$.

Proof. The LHS of the inequality monotonically increase wrt x , while the RHS monotonically decrease wrt x . Hence the solution set is in the form $(T_n, 1]$. $\square$

We can also say something about T_n :

Theorem 2. The optimal thresholding sequence T_n has the following properties:

1. $T_0 = 0$ and $T_1 = \frac{1}{2}$
2. T_n strictly monotonically increase.
3. $\lim_{n \rightarrow \infty} T_n = 1$

Proof. TBD. $\square$

Now we examine the asymptotic behavior of the optimal policy.

Theorem 3. The optimal policy ensures a $\frac{1}{2}$ winning rate, and the bound is tight for large n . In other words, $V(0, n, 0) \downarrow \frac{1}{2}$.

Proof. TBD. $\square$

Numerical result for T_n is summarized in the following table.

4 Relaxed Problem with Error

Fix $\delta > 0$, and use the following notation for game state: $[\tilde{x}, n]$ means the current item to be decided has observed (inaccurate) value $\tilde{x}$, and we have n items after the current item. If the player examines the current item, the game state evolves into $[x, n-1]$, where we now have the exact value x , and one item (the next) is omitted. If the player otherwise declines the current item, the game state evolves into $[\tilde{x}_n, n-1]$ where $\tilde{x}$ is replaced by $\tilde{x}_n$, the next item, and we have one item less in the queue.

Table 1: Numerical Result for T_n

n	T_n	$EV(\cdot, n, 0)$
0	0.	1.
1	.500	.750
2	.690	.684
3	.776	.655
4	.825	.639
5	.856	.629
6	.878	.622
7	.894	.616
8	.906	.612
9	.916	.609
10	.924	.606

We again obtain the DP equations. $a \wedge b$ denotes $\max(a, b)$.

$$V(x, 0, x^*) = \mathbf{1}(x > x^*) \quad (17)$$

$$V(\tilde{x}, n, x^*) = \max \left(EV(X, n-1, x^* \wedge X_{n-1}), EV(\tilde{X}_{n-1}, n-1, x^* \wedge X) \right) \quad (18)$$

$$V(x, n, x^*) = \begin{cases} \max \left(x^n, E \left[V(\tilde{X}_{n-1}, n-1, x) \right] \right) & x \geq x^* \\ E \left[V(\tilde{X}_{n-1}, n-1, x^*) \right] & x < x^* \end{cases} \quad (19)$$

The problem with error is again solvable using thresholding:

Theorem 4. The optimal policy for the problem with error can be described completely by thresholding. In other words, for each $\delta \geq 0$, there exists sequences T_n and $R_n \in \mathbb{R}$ such that the LHS in the max operator of (18) is greater than the RHS iff $\tilde{x} > R_n$. Similarly, the LHS is picked in (19) iff $x > T_n$.

Proof. (incomplete) We first prove the existence of T_n . In (19), note that $V(\tilde{X}_{n-1}, n-1, x)$ is monotone decreasing wrt x for each specific realization of $\tilde{X}_{n-1}$. Taking expectation won't affect monotonicity, so the argument in the previous theorem applies.

Now we work on existence of R_n and consider (18). Let $A(x) := E[V(x, n-1, x^* \wedge X_{n-1})]$ and $B(x) := E[V(\tilde{X}_{n-1}, n-1, x^* \wedge x)]$, where the expectation is taken over $(u - \delta, u + \delta) \cap (0, 1)$. Our goal is to prove that $EA > EB$ is a thresholding on u . We will first show that $A(x) > B(x)$ is a thresholding on x , and apply properties of expectation.

Let $x < x^*$. Now $A(x) = E \left[V \left(\tilde{X}_{n-2}, n-2, x^* \wedge X_{n-1} \right) \right]$ and $B(x) = E \left[V \left(\tilde{X}_{n-1}, n-1, x^* \right) \right]$. $B(x)$ is measure of the event that the best item is selected with value $> x^*$, while $A(x)$ is measure of the same event with additional constraint that X_{n-1} is not the best item. Hence $A(x)$ is a sub-event of $B(x)$, and $A(x) < B(x)$.

Let $x > x^*$ and assume $x < T_n$. Now $A(x) = \mathbb{E} \left[V \left(\tilde{X}_{n-2}, n-2, x \wedge X_{n-1} \right) \right]$ and $B(x) = \mathbb{E} \left[V \left(\tilde{X}_{n-1}, n-1, x \right) \right]$. By the same argument, $A(x) < B(x)$.

Now assume $x > T_n$. Take $\omega \in \Omega$. For all $x < X_{n-1}(\omega)$, we have $A_\omega(x) < B(x)$ (why? Need to clarify). For all $x > X_{n-1}(\omega)$, $A_\omega(x) = x^{n-1}$ monotonically increase up to $A(1) = 1$. Hence we have a unique intersection $\alpha \in (T_n, 1)$ where $x > \alpha \Leftrightarrow A_\omega(x) > B(x)$. α does not depend on ω , so it's the unique intersection between $B(x)$ and the "averaged" $A(x)$.

(Another way to proceed) Now assume $x > T_n$. Evaluate the expectation over X_{n-1} , we have

$$A(x) = \int_0^x x^{n-1} dy + \int_x^1 \mathbb{E} V(\tilde{X}_{n-2}, n-2, y) dy.$$

As x increases from T_n to 1, the first term monotonically increase from T_n^n to 1, while the second term monotonically decrease and vanishes. $A(x)$ is continuous, so there's at least one intersection. (remains to show uniqueness) $\square$

Again we can say something about the thresholding sequences.

Theorem 5. The optimal thresholding sequences T_n and R_n has the following properties:

1. When $\delta = 0$, $T_n \equiv R_n$ are same as the problem without error.
2. For fixed δ , T_n and R_n both strictly monotonically increase wrt n .
3. For fixed n , T_n and R_n both monotonically decrease wrt δ .
4. $\lim_{n \rightarrow \infty} T_n = 1$
5. (?) $\lim_{n \rightarrow \infty} R_n = 1 - \delta$.
6. As $\delta \rightarrow \infty$, $R_n^{(\delta)} \rightarrow -\infty$ and $T_n^{(\delta)} \rightarrow T_{\lfloor \frac{n}{2} \rfloor}^{(0)}$ for all n .

Proof. TBD. $\square$

Corollary 1. The optimal winning rate $V(0, n)$ satisfies the following:

1. $V_{(0)}$ is identical to the problem without error.
2. As $\delta \rightarrow \infty$, $V_{(\delta)}(0, n) \rightarrow \frac{1}{2} V_{(0)}(0, \lceil \frac{n}{2} \rceil)$ for all n .

We can compute the optimal thresholds by iteratively solving inequalities.

Lecture 6

Thu 21 Apr 2022 00:09

5 Noisy SP with examine

5.1 Problem Construction from R.V.s

This is partly inspired by (Krieger and Samuel-Cahn, 2012). Consider a set of items $P = \{x_1, \dots, x_n\}$. Let X_i be i.i.d. continuous R.V.s, and $Y_i = X_i + \varepsilon_i$, and the ε_i are i.i.d. noise variables. The goal is to maximize the probability of selecting the i for which the X value is maximal.

Upon arrival of each item x_i , the player may choose to either reject the item, or to examine this item, forfeiting k items. After examination the player must decide whether to accept or reject it. Hence if $t \geq n - k|S|$ where S is the set of examined items, the game terminates and the current item at hand is selected. We denote the number of remaining items by $n_t = n - k|S_t|$.

Upon each arrival the player observes a total order of $P_t = \{x_1, \dots, x_t\}$ sorted by their corresponding Y_i , and for each *examined* item x_i , Y_i replaced by X_i .

We may limit the number of examinations throughout the game by m .

1. When $m = 0$, the game is Noisy SP.
2. When $k = 0$, this is a Noisy SP with limited query from an oracle.
3. When $k = 0$ and $m \geq n$, this is Classical SP.

Claim. The optimal strategy can always achieve an asymptotic winning rate $\frac{1}{(1+k)e} + \delta$ for some $\delta > 0$.

Conjecture. A nontrivial lower bound can be obtained with $m = o(n)$, possibly $O(\log n)$ examinations.

Conjecture. The optimal solution is in the following form: there exists $1 \leq k_1 \leq k_2 \leq \dots \leq k_n \leq n$ and $1 \leq R \leq n$, and for each arriving item x_i , examine if its relative rank $\leq k_{i+n-n_t}$, and decline otherwise. After examination, accept if it has relative rank 1, and $i > R$.

5.2 Problem Construction from permutation group

Let S_n be the permutation group on $[n]$. We identify the *swaps* $\tau_i : i \leftrightarrow i + 1$, $i = 1 \dots n - 1$. Define i.i.d. poisson R.V.s $N_i \sim \text{Po}(\lambda)$ to be the *multiplicity* of τ_i .

Now we randomly permute $\tau_i, \dots, \tau_{n-1}$ where each τ_i appears N_i times. We denote the resultant array by $\tau_{(j)}, j = 1 \dots N = \sum N_i$. Define the *resultant permutation* π by

$$\pi = \tau_{(1)} \circ \tau_{(2)} \circ \dots \circ \tau_{(N)}.$$

Claim. $\mathbb{P}(\pi) = f(d(\pi, I))$ where I is identity permutation, and d is *Kendall-tau Distance*. Moreover, f is strictly monotone decreasing.

Conjecture. π is distributed as if we observe the effect of ε_i on X_i , for arbitrary continuous i.i.d. distribution, under some additional conditions.

The player observes π_t which is induced by π on $[t]$. The distribution of π is restricted according to the set of examined items E_t such that

1. $i, j \in E_t \Rightarrow (\pi(i) < \pi(j) \Leftrightarrow i < j)$
2. some other conditions that uniquely determine the restriction.

Claim. The optimal winning rate, $W_n(\lambda)$ goes to $\frac{1}{(k+1)^e}$ as $\lambda \rightarrow \infty$, where the distribution of π becomes uniform in S_n .

Lecture 7: Formulation for Noisy SP with examination

Tue 03 May 2022 05:45

Let S_i be set of examined items at round i . Let Z_i be the rank affected by examination, that is

$$Z_i = \begin{cases} X_i & i \in S_i \\ Y_i & i \notin S_i \end{cases}.$$

Define the absolute and relative ranks by

$$\text{AR}(X_i) := \sum_{j=1}^n \mathbf{1}(X_i \leq X_j) \quad (20)$$

$$\text{RR}(X_i) := \sum_{j=1}^i \mathbf{1}(X_i \leq X_j) \quad (21)$$

$$(22)$$

Also define $\text{AR}(Y_i)$, $\text{AR}(Z_i)$, $\text{RR}(Y_i)$, and $\text{RR}(Z_i)$ in similar manner.

Let $\mathcal{F}_i := \sigma(\text{RR}(Y_1), \dots, \text{RR}(Y_i), \text{RR}(Z_1), \text{RR}(Z_{i-1}))$.

The examined objects is a random set $S_i \in \mathcal{F}_i$ such that $S_0 = \emptyset$, $S_i \subset S_{i+1} \subset S_i \cup \{i+1\}$ and $|S_n| \leq m$.

The decision to accept is a stopping time $\{\tau = i\} \in \sigma(\mathcal{F}_i, \text{RR}(Z_i))$.

Lecture 8: Partially Adversarial SP with Examination

Tue 31 May 2022 08:47

6 Partially Adversarial SP

6.1 Problem formulation

Random arrival. n items are totally ordered. A random permutation σ is taken uniformly from the permutation group S_n . The classical SP corresponds to finding a stopping time that most likely stops on $t = \sigma(1)$. We will always use numbers to denote the true rank of items, so the incoming stream of items is represented by $\sigma(1, \dots, n)$.

Adversarial noise. The adversary adds a random noise with resultant permutation π . This permutation is not over the arriving order of object, but over the observed rank, i.e. $\pi(j) = i$ means that the item with actual rank j will be found by the player to be i -th best, if the player observes all items without examining them.

Restriction on adversary. To restrict the power of adversary, we demand π to be long-range stable, i.e. $\pi(i) < \pi(j)$ whenever $i < j$ and $j - i > c$, where the constant c is known to the algorithm.

Examination. For each incoming item the player can choose from three options: Accept, Decline, and Examine. If the player chooses to Examine, the current item will be added to a set E of examined items, and the next item will be forfeited and left unobserved. We use E_t to denote the items examined by the start of t -th round. The Examine option is not available if the current item is the last item.

Observable quantities. Let O_t be the observed total order on round t . It is a total order on $[t]$ induced by $\pi\sigma^{-1}$. Let $\hat{O}_t$ be the actual total order on round t . It is a total order on $[t]$ induced by σ^{-1} . Let P_t be the total order obtained from examination, defined by restricting $\hat{O}_t$ to E_t . Note that O_t and P_t may not be compatible since O_t is subject to observational "mistakes" caused by the noise. We use $\geq_{O_t}$ to denote the order inside the ordered set O_t .

It is equivalent, but sometimes more convenient, to use the notion of relative rank instead of total order. We define $\text{RR}_t(i) = \sum_{s=1}^t \mathbf{1}(\pi\sigma^{-1}(s) \geq \pi\sigma^{-1}(i))$ to be the observed relative rank function.

Decision rule. Let $(\mathcal{F}_t)_{t \in [n]}$ be the sequence of σ -algebras with each $\mathcal{F}_t$ generated by the random variables $O_1 \times P_1, \dots, O_t \times P_t$, that is,

$$\mathcal{F}_t = \sigma(O_1 \times P_1, \dots, O_t \times P_t) = \sigma(O_t \times P_t).$$

By a stopping time, we mean a random variable τ taking values in $[n]$ and satisfying the property $\{\tau = t\} \in \mathcal{F}_t$. The decision rule is a stopping time τ together with a sequence of indicators $\text{Exm}_t := \mathbf{1}(t \in E_{t+1})$ such that

$$\text{Exm}_{t+1} \in \mathcal{F}_t \text{ and } \text{Exm}_t \text{Exm}_{t+1} = 0 \text{ and } \mathbf{1}(\tau = t) \text{Exm}_t = 0.$$

Goal. Our goal is to choose a decision rule (τ, Exm) to maximize $\Pr[\tau = \sigma(1)]$.

6.2 Preliminaries

We list several obvious facts about this problem construction.

- Lemma 1.** 1. The relative ranks behave the same: $RR(t)$ is uniformly sampled from $1, \dots, t+1$ and independent from each other.
2. When π do not depend on σ , $\sigma\pi$ and σ have exactly same distribution.

6.3 Case without examination

We first do not allow examination, and see how far the algorithm can get.

6.3.1 Weak adversarial, do not know anything

Consider the weakest adversary, that can only pick a single π without knowing about the algorithm. The competitive ratio is

$$\begin{aligned} & \inf_{\pi} \sup_{\tau} \mathbb{E}_{\sigma} [\Pr [\tau = \sigma\pi(1)]] \\ &= \inf_k \inf_{\pi: \pi(1)=k} \sup_{\tau} \mathbb{E}_{\sigma} [\Pr [\tau = \sigma(k)]] \\ &\leq \inf_k \sup_{\tau} \mathbb{E}_{\sigma} [\Pr [\tau = \sigma(k)]] . \end{aligned}$$

The problem reduces to a generalized post-doc variant of SP, where we want the item with absolute rank k instead of 1. Each time k is provided, we may specifically design our algorithm to maximize the probability of choosing the rank k item. This is significantly harder to solve as compared to the classical SP, in its value function is no longer monotone: the function

$$\Pr [\text{AR}(t) = m | \text{RR}(t) = r] = \frac{\binom{m-1}{r-1} \binom{n-m}{t-r}}{\binom{n}{t}}$$

is not monotone wrt r for any fixed m . No explicit solution has been found so far. We can show that the worst case when $c < \frac{n}{2}$ is attained at $\pi(1) = c$, and at $\pi(1) = \lfloor \frac{n}{2} \rfloor$ when $c \geq \frac{n}{2}$.

6.3.2 Weak adversarial, know algorithm

Now we assume that the adversary adapts to each algorithm, but cannot depend on the random permutation σ . The competitive ratio is now

$$\sup_{\tau} \inf_{\pi} \mathbb{E}_{\sigma} [\Pr [\tau = \sigma\pi(1)]]$$

Note that the generalized postdoc variant of SP is an upper bound to the problem:

$$\begin{aligned} &= \sup_{\tau} \inf_k \inf_{\pi: \pi(1)=k} \mathbb{E}_{\sigma} \Pr [\tau = \sigma(k)] \\ &\leq \sup_{\tau} \inf_k \mathbb{E}_{\sigma} \Pr [\tau = \sigma(k)] . \end{aligned}$$

This time we need to tackle all sub-problems using only one algorithm. In other words, the algorithm's probability to choose the 1, 2, ..., c -th item are compared, and the minimum one is the CR.

For lower bound, we consider the following algorithm: When each item arrives, if $RR(t) > c$, reject the item. Otherwise, accept with probability p , where p is a tuning parameter. Now for some algorithm τ and fixed k , We derive analytic result for this algorithm. Its performance monotonously decrease wrt $\pi(1)$ for fixed n and p , and always approach the trivial lower bound $\frac{1}{n}$.

Consider another algorithm: Let the first pn items to go by, and then select the first item with relative rank better or equal to c . This algorithm makes no distinction among the first c items, so it suffices to analyze its behavior on the classic SP.

6.3.3 Strong adversarial, knows τ and σ

$$\sup_{\tau} \mathbb{E}_{\sigma} \inf_{\pi} [\Pr [\tau = \sigma\pi(1)]] .$$

6.3.4 Strong adversarial, knows τ , σ and $\mathbf{r}$

$$\sup_{\tau} \mathbb{E}_{\sigma} \left[\mathbb{E}_{\mathbf{r}} \left[\inf_{\pi} \mathbf{1} (\tau = \sigma\pi(1)) \right] \right] .$$

In this case the adversary knows the random seed used by the algorithm, so it's able to make hindsight decisions. In this setting, the algorithm achieve zero cr. This can be shown by calling to an auxiliary problem of adversarial best-choice within the top c items, where no deterministic algorithm can guarantee nonzero competitive ratio against an adversary.

6.4 Lower Bound for an algorithm

We establish a lower bound for the following simple algorithm: examine each item with $RR_t(t) \leq c$. Accept the current item if $t \geq T$ and the item is the best so far.

Algorithm 3 Simple algorithm

```

1: while At least two items are left do
2:    $x \leftarrow$  the current item to be decided
3:    $t, O_t, P_t \leftarrow$  as defined before
4:   if  $x \in P_t$  then
5:     if  $x = \max P_t$  and  $t \geq T$  then
6:       Accept  $x$ 
7:     else
8:       Decline  $x$ 
9:     end if
10:  else if  $|\{a \in O_t : a \geq_{O_t} x\}| \leq c$  then  $\triangleright$  Means  $RR_t(x) \leq c$ 
11:    Examine  $x$ 
12:  else
13:    Decline  $x$ 
14:  end if
15: end while
16: Accept the only item left

```

Lemma 2 (Equivalent construction of the game). It is equivalent to view this game as follows: when an item is examined, instead of omitting the next item, the last available item is taken from the end of the queue. The player wins only if the best item is accepted, and the item is not omitted.

Proof. (informal, to be put rigorously) Consider a deterministic algorithm. The algorithm never knows $\sigma^{-1}(t)$, the position of the current item in the arrival sequence. For some $\sigma \in S_n$, we can perform some reordering according to the following rule: We first let the algorithm run through the sequence, and take out all the items omitted, and stack them to the end of the queue. The resultant sequence σ' looks exactly same to the algorithm in the new setting. Hence, the whole gameplay will be identical up to an automorphism on S_n , and there's no difference revealed to the algorithm. Because the only randomness we used is the randomized $\sigma \in S_n$, we have essentially created an isomorphism between the two probability spaces for the two versions of the game. $\square$

Intuition. Imagine a strip of paper which represents the incoming sequence of items. One consumes the items from left to right, but the paper strip burns from right to left (due to examination which makes some items unavailable). The moment the paper is used up, the game is forced to end.

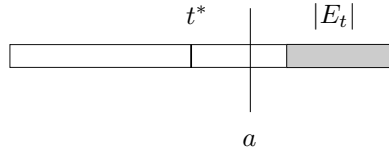


Figure 1: Our scheme to finding a lower bound

Let $t^* := \sigma^{-1}(1)$, i.e., the position of the best item. We couple the game with the classical secretary problem (CSP).

Lemma 3 (Coupling with CSP). Whenever an algorithm wins the problem, it automatically wins CSP in the same item setting. Whenever an algorithm wins CSP such that $t^* + |E_{t^*}| \leq n$, it automatically wins our problem.

Proof. To be done. $\square$

Lemma 4. For any a and $n > a$,

$$\begin{aligned} \Pr[\text{win CSP with } a \text{ items } |t^* + |E_{t^*}| \leq n] \\ \geq \Pr[\text{win CSP with } a \text{ items}] \end{aligned}$$

Proof. To be done. $\square$

From Lemma 3 we have

$$\Pr[\text{win}] = \Pr[\text{win CSP with } n \text{ items and } t^* + |E_{t^*}| \leq n] \quad (23)$$

Now we constrain t^* and $|E_{t^*}|$ by a to make sure they don't meet.

$$\geq \max_{a \leq n} \Pr[\text{win CSP with } n \text{ items and } t^* + |E_{t^*}| \leq n \cap t^* \leq a] \quad (24)$$

Which follows from property of intersection. Continue to apply elementary properties,

$$= \max_{a \leq n} \Pr[\text{win CSP with } n \text{ items } |t^* + |E_{t^*}| \leq n \cap t^* \leq a] \quad (25)$$

$$\Pr[t^* + |E_{t^*}| \leq n | t^* \leq a] \Pr[t^* \leq a]$$

$$= \max_{a \leq n} \Pr[\text{win CSP with } n \text{ items } |t^* + |E_{t^*}| \leq n \cap t^* \leq a] \quad (26)$$

$$\Pr[|E_{t^*}| \leq n - a] \Pr[t^* \leq a]$$

Since a CSP with n items where only the first a items may be the best is identical to a CSP with a items,

$$= \max_{a \leq n} \Pr[\text{win CSP with } a \text{ items } |t^* + |E_{t^*}| \leq n] \quad (27)$$

$$\Pr[|E_{t^*}| \leq n - a] \Pr[t^* \leq a]$$

By Lemma 4

$$= \max_{a \leq n} \Pr[\text{win CSP with } a \text{ items}] \quad (28)$$

$$\Pr[|E_{t^*}| \leq n - a] \Pr[t^* \leq a]$$

Evaluate each term separately,

$$\Pr[\text{win}] \geq \max_{a \leq n} \frac{1}{e} \Pr[|E_a| \leq n - a] \Pr[t^* \leq a]. \quad (29)$$

For $t \geq c$ we have

$$|E_t| = c - 1 + c \sum_{s=c}^t B_s$$

where $B_s \sim \text{Ber}(\frac{1}{s})$.

Using Hoeffding's Bound:

$$\Pr\left[\sum_c^t B_s - \sum_c^t \frac{1}{s} \leq \beta\right] \geq 1 - \exp\left(-\frac{2\beta^2}{t - c + 1}\right)$$

$$\Pr \left[|E_t| \leq c(\beta + \sum_{s=c}^t \frac{1}{s}) + c - 1 \right] \geq 1 - \exp \left(-\frac{2\beta^2}{t - c + 1} \right).$$

Fix some a . Let β be the greatest real number such that

$$c(\beta + \sum_{s=c}^a \frac{1}{s}) + c - 1 \leq n - a.$$

Therefore,

$$\beta = \frac{n - a}{c} - 1 - H(a) + H(c - 1)$$

where we use $H(a)$ to denote the a -th harmonic number.

Now we have a concentration bound for $|E_{t^*}|$:

$$\Pr [|E_{t^*}| \leq n - a] \geq 1 - \exp \left(-\frac{2\beta^2}{t - c + 1} \right).$$

Applying this bound, we now have

$$\Pr [\text{win}] \geq \max_{a \leq n} \frac{1}{e} \frac{a}{n} \left(1 - \exp \left(-\frac{2\beta^2}{a - c + 1} \right) \right).$$

Numerical Result. Take $c = 2$ for an example. When $a = 70$, we have the numerical bound $\Pr [\text{win}] \geq 0.674 \frac{1}{e}$, which is a non-trivial improvement from the examine-everything algorithm which guarantees only a competitive ratio of $0.5 \frac{1}{e}$.

6.5 Upperbound for the problem

This section only contains ideas, there's not any proof.

6.5.1 some guesses

Definition 2. A *simple algorithm* is an algorithm in the following form, with parameters c and T replaced by any valid value: **Examine** each item with observed relative rank $\leq c$, and **Accept** if it is $\max P_{t+1}$ and $t \geq T$.

Conjecture 1. Any algorithm, no matter deterministic or randomized, has its competitive ratio dominated by some simple algorithm.

Conjecture 2. For each c and n (as parameters of the game), of all simple best algorithms, there is one optimal simple algorithm. The optimal algorithm has the exactly same c .

If the above guesses are true, then we only need to find the optimal T for the simple algorithm, and bound its competitive ratio from above.

6.5.2 An alternative thought

The process of examination and omitting items will often bring the player to a **from-two-choose-one** scenario. For example, consider the following sequence of items:

$*, \triangleleft, \triangleright, -, -$

Where the player is at $*$. Suppose that both $\triangleleft$ and $\triangleright$ are "critical" items such that they have relative rank $\leq c$ and shall be examined. Now if the player declines $*$, it has to examine $\triangleleft$ and therefore lose $\triangleright$; if it examines and declines $*$, it loses $\triangleleft$.

Therefore one of $\triangleleft$ and $\triangleright$ must be lost, and there is no way to mitigate this. This behavior can be something fundamental that can be properly exploited to obtain an upper bound.

References

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